

Week 1: Whole Numbers

JSS1 Mathematics — First Term Lesson Note (Deep-Dive Edition)

Introduction

Numbers are everywhere in daily life — counting money, reading a phone number, checking a population figure, or reading the time. Before we can work confidently with any of these, we need a solid understanding of **whole numbers**, the basic building blocks of arithmetic.

Definition

Whole numbers are the set of numbers: **0, 1, 2, 3, 4, 5, ...** — counting numbers together with zero. They have no fractions or decimals and are never negative.

Note: Natural numbers (also called counting numbers) are 1, 2, 3, 4... — the same as whole numbers but **without zero**. Every natural number is a whole number, but 0 is a whole number that is not a natural number.

Place Value

Every digit in a number has a **place value** depending on its position. For example, in the number **4,732,619**:

Digit	4	7	3	2	6	1	9
Place	Millions	Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Units
Value	4,000,000	700,000	30,000	2,000	600	10	9

Expanded form: $4,732,619 = 4,000,000 + 700,000 + 30,000 + 2,000 + 600 + 10 + 9$

Second Example: Write 305,208 in expanded form.

$305,208 = 300,000 + 5,000 + 200 + 8$ (note: there are no ten-thousands or tens, so those places are skipped)

Ordering Whole Numbers

We compare numbers using the symbols: $>$ greater than, $<$ less than, $=$ equal to.

Example 1: Arrange 3,204; 320; 32,040 in ascending order (smallest to biggest).

→ $320 < 3,204 < 32,040$

Example 2: Arrange 15,600; 15,060; 15,006 in descending order (biggest to smallest).

→ $15,600 > 15,060 > 15,006$

Properties of Whole Numbers

1. **Closure Property** — Adding or multiplying two whole numbers always gives a whole number. ($5 + 3 = 8$ ✓; $4 \times 6 = 24$ ✓)

2. **Commutative Property** — Order doesn't matter for addition/multiplication. ($4 + 6 = 6 + 4$; $3 \times 5 = 5 \times 3$)
3. **Associative Property** — Grouping doesn't matter. $((2 + 3) + 4 = 2 + (3 + 4) = 9)$
4. **Distributive Property** — $a \times (b + c) = (a \times b) + (a \times c)$. Example: $3 \times (4 + 5) = (3 \times 4) + (3 \times 5) = 27$
5. **Identity Property** — Adding 0 or multiplying by 1 leaves a number unchanged. ($7 + 0 = 7$; $9 \times 1 = 9$)

Real-Life Application

Whole numbers are used whenever we count things that cannot be split into parts — the number of students in a class, the number of houses on a street, or the population of a town. A market trader counting 250 tubers of yam, or a school recording 45 students in JSS1A, is using whole numbers directly.

Vocabulary

Word	Meaning
Digit	Any of the symbols 0–9 used to write numbers
Numeral	A symbol or group of symbols representing a number
Ascending order	Arrangement from smallest to biggest
Descending order	Arrangement from biggest to smallest

Evaluation Questions

- 1 Write 6,082,451 in expanded form.
- 2 Arrange in descending order: 8,901; 890; 89,010; 8,090
- 3 State the property shown: $6 \times (2 + 7) = (6 \times 2) + (6 \times 7)$
- 4 What is the place value of 5 in 251,637?
- 5 Explain the difference between a whole number and a natural number.
- 6 Is 0 a whole number? Is 0 a natural number? Explain.
- 7 Write the number 720,045 in expanded form.
- 8 Arrange in ascending order: 4,502; 450; 45,020; 4,052
- 9 Give a real-life example of when you would use whole numbers to count something.
- 10 State and explain the identity property of whole numbers using an example of your own.